

Analog IC Design

Lecture 23 Reference Circuits

Dr. Hesham A. Omran

Integrated Circuits Laboratory (ICL)
Electronics and Communications Eng. Dept.
Faculty of Engineering
Ain Shams University

Outline

- ❑ Reference circuits
- ❑ Bandgap reference (BGR) circuit
 - BGR basic operation
 - BGR practical CMOS implementation
- ❑ Constant-gm circuit

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Reference Circuits

- ❑ Stable DC voltage and DC current generation
 - Little dependence on process (P) and supply (V)
 - Well-defined dependence on temperature (T)
 - Not necessarily independent!
- ❑ Temperature dependence
 - Positive temperature coefficient (+ve TC): Proportional to absolute temperature (PTAT)
 - Negative temperature coefficient (-ve TC): Complementary to absolute temperature (CTAT)
 - **Temperature independent (Zero-TC): $ZTAT = PTAT + CTAT$**
- ❑ Most process parameters vary with temperature
 - If we achieve a temperature independent reference it will also be process independent

Why Reference Circuits?

- ❑ Applications:
 - Voltage regulators
 - ADCs and DACs
 - Biasing of amplifiers
 - Common-mode (CM) level of fully-differential circuits
 - ...

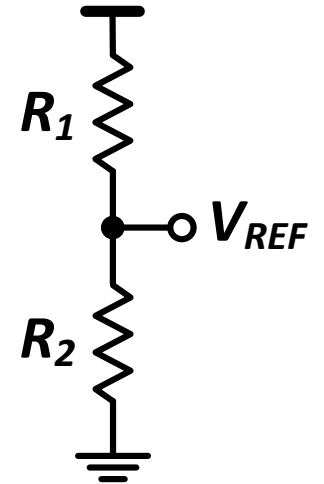
- ❑ Types of references we usually need (all should be PV insensitive):
 - PTAT: Proportional to absolute temperature (+ve TC)
 - Temperature independent (Zero-TC) → PVT insensitive
 - Constant-transconductance (constant-Gm)

Poor Man's Reference Voltage

$$V_{REF} = V_{DD} \frac{R_2}{R_1 + R_2}$$

- ❑ Good layout makes V_{REF} independent of process and temperature.
- ❑ V_{REF} absolute value depends on V_{DD} .
- ❑ V_{REF} is sensitive to V_{DD} variations (poor PSR).

$$PSR = \frac{\Delta V_{REF}}{\Delta V_{DD}} = \frac{R_2}{R_1 + R_2}$$



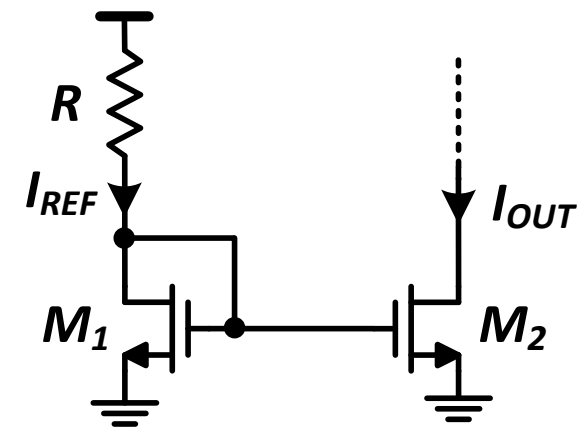
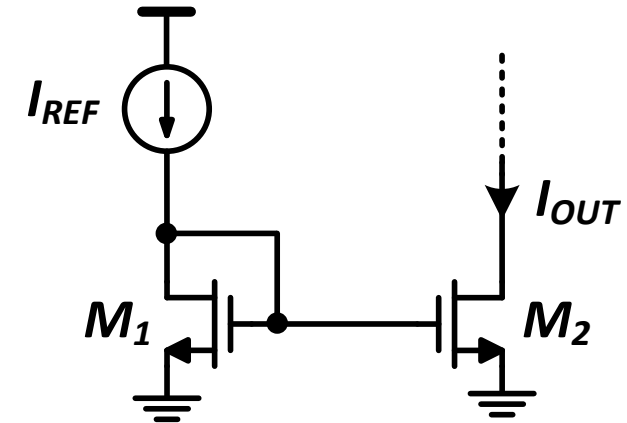
Poor Man's Reference Current

$$I_{REF} = \frac{\mu C_{ox}}{2} \frac{W_1}{L_1} (V_{GS} - V_{TH})^2 = \frac{\beta_1}{2} (V_{GS} - V_{TH})^2$$

$$I_{REF} = \frac{V_{DD} - V_{GS}}{R} = \frac{V_{DD} - \sqrt{\frac{2I_{REF}}{\beta_1}} - V_{TH}}{R}$$

- ❑ I_{REF} depends on process, supply voltage, and temperature (PVT).
- ❑ I_{REF} is sensitive to V_{DD} variations (poor PSR).

$$\Delta I_{REF} = \frac{\Delta V_{DD}}{R + 1/g_{m1}} \rightarrow \frac{\Delta I_{REF}}{\Delta V_{DD}} \approx \frac{1}{R}$$

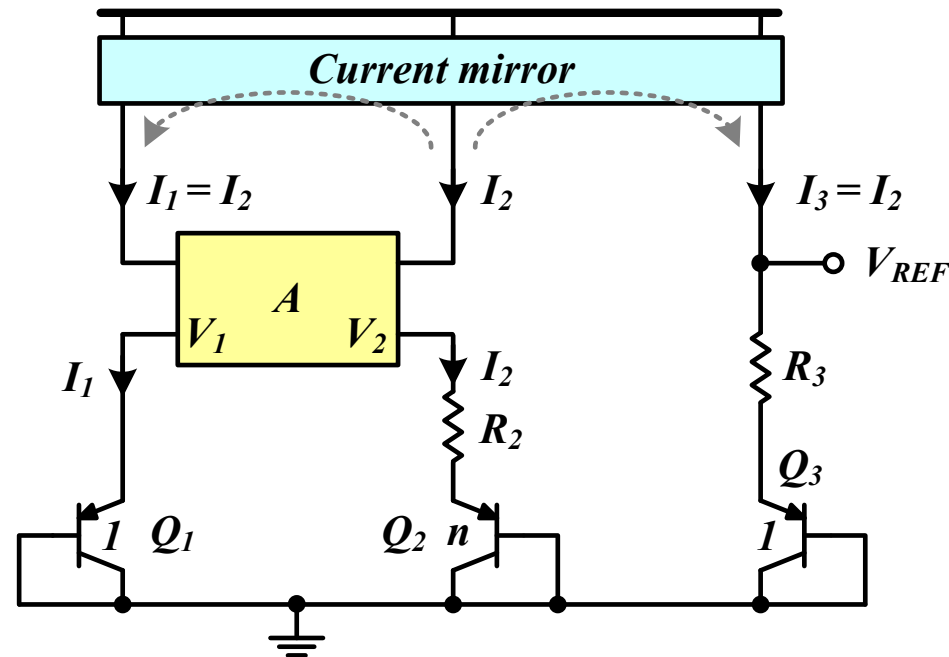


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Basic Bandgap Reference (BGR) Circuit

- ❑ Q1, Q2, and Q3 are diode connected PNP BJTs.
- ❑ The A-block is a circuit that makes $V_1 = V_2$.
- ❑ The current mirror copies I_2 to I_1 and I_3 .
- ❑ V_{REF} is ZTAT $\rightarrow$ How?



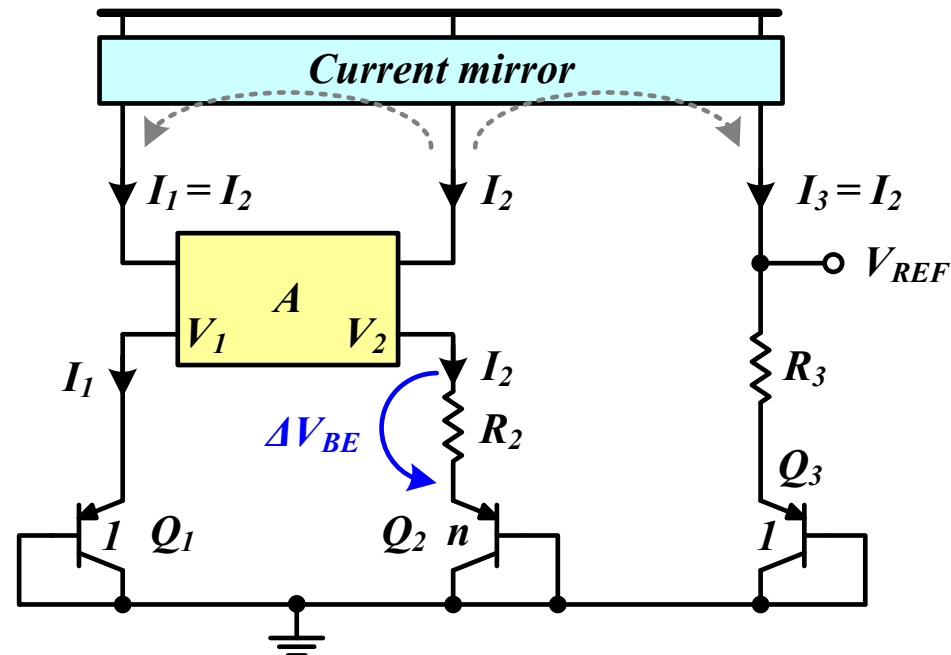
How much is I_2 ?

❑ Using simple diode model you may expect $I_2 = 0$.

❑ But Q2 is n BJTs (diodes) connected in parallel.

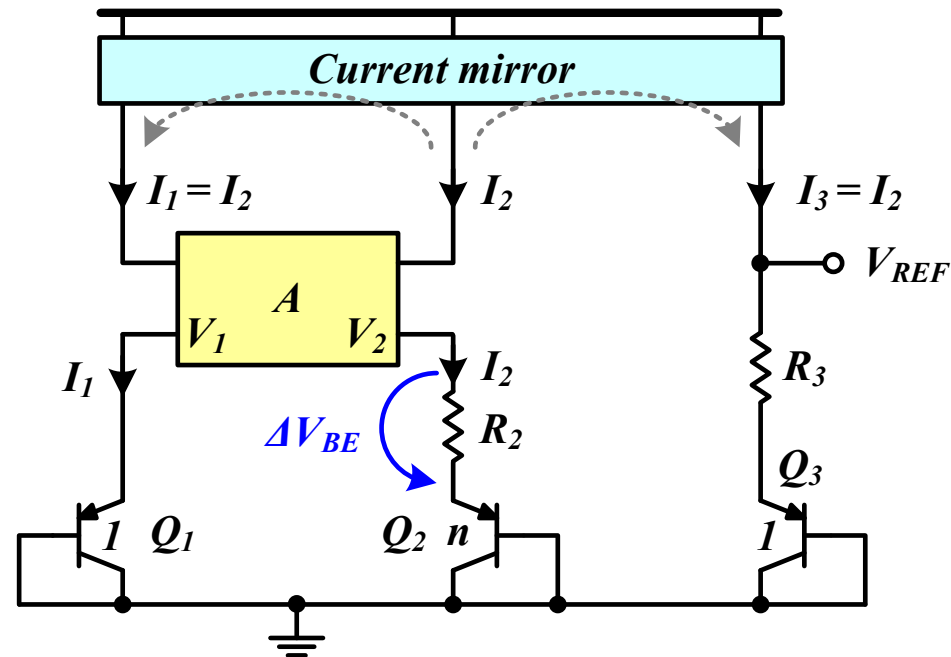
▪ $I_{C1} \approx I_1$ and $I_{C2} \approx I_2/n$

❑ $I_C = I_S e^{\frac{|V_{BE}|}{V_T}} \rightarrow |V_{BE}| = V_T \ln \frac{I_C}{I_S}$



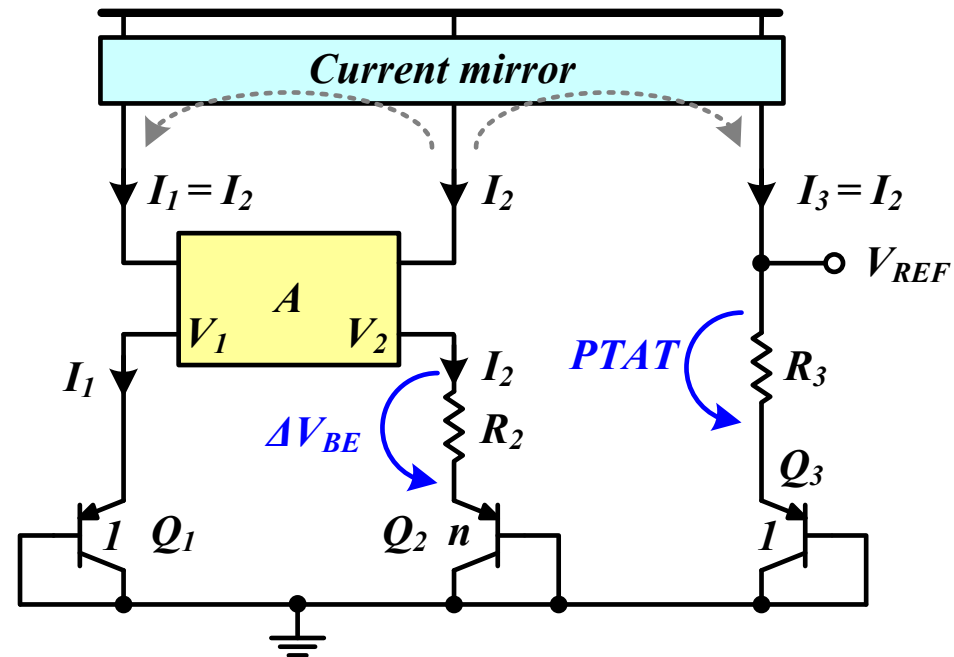
I_2 is PTAT

- $I_C = I_S e^{\frac{|V_{BE}|}{V_T}} \rightarrow |V_{BE}| = V_T \ln \frac{I_C}{I_S}$
- $I_{C1} \approx I_1$ and $I_{C2} \approx I_2/n$
- $|V_{BE1}| \approx V_T \ln \frac{I_1}{I_S}$ and $|V_{BE2}| \approx V_T \ln \frac{I_2/n}{I_S} = V_T \ln \frac{I_2}{I_S} - V_T \ln n$
- $\Delta V_{BE} = |V_{BE1}| - |V_{BE2}| = V_T \ln n = \frac{kT}{q} \ln n \propto T \rightarrow \text{PTAT!}$



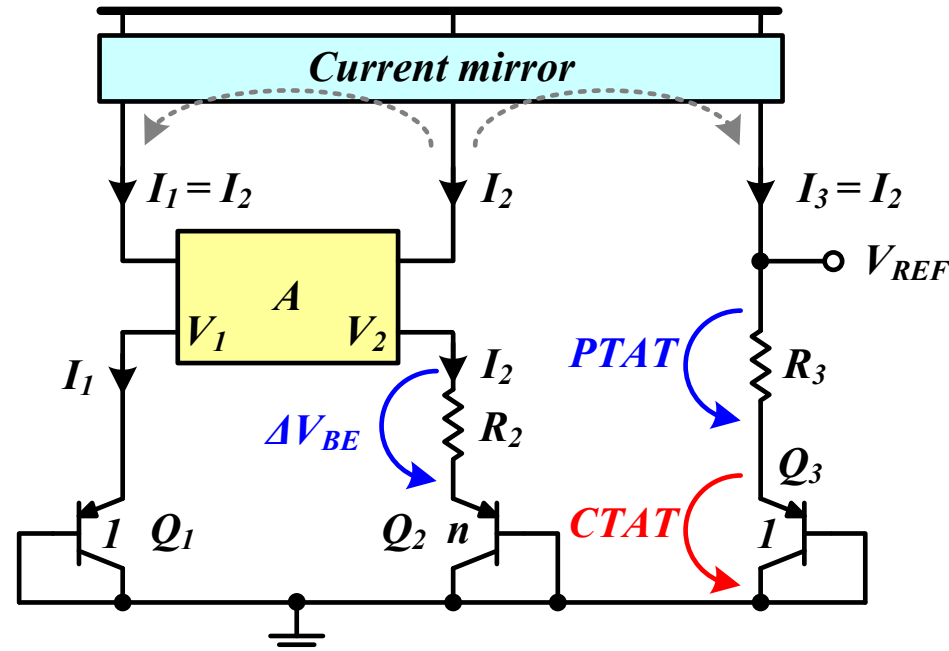
The PTAT Component

- $\Delta V_{BE} = |V_{BE1}| - |V_{BE2}| = V_T \ln n = \frac{kT}{q} \ln n \propto T \rightarrow \text{PTAT}$
- $I_2 = \frac{\Delta V_{BE}}{R_2} = \frac{kT}{q} \ln n \cdot \frac{1}{R_2} \propto T \rightarrow \text{PTAT}$
- $V_{R_3} = I_3 R_3 = \Delta V_{BE} \cdot \frac{R_3}{R_2} = V_T \ln n \cdot \frac{R_3}{R_2} = a_1 T \propto T \rightarrow \text{PTAT}$
- $a_1 = \frac{k}{q} \ln n \cdot \frac{R_3}{R_2} \approx 0.086 \ln n \cdot \frac{R_3}{R_2} \text{ mV/K}$



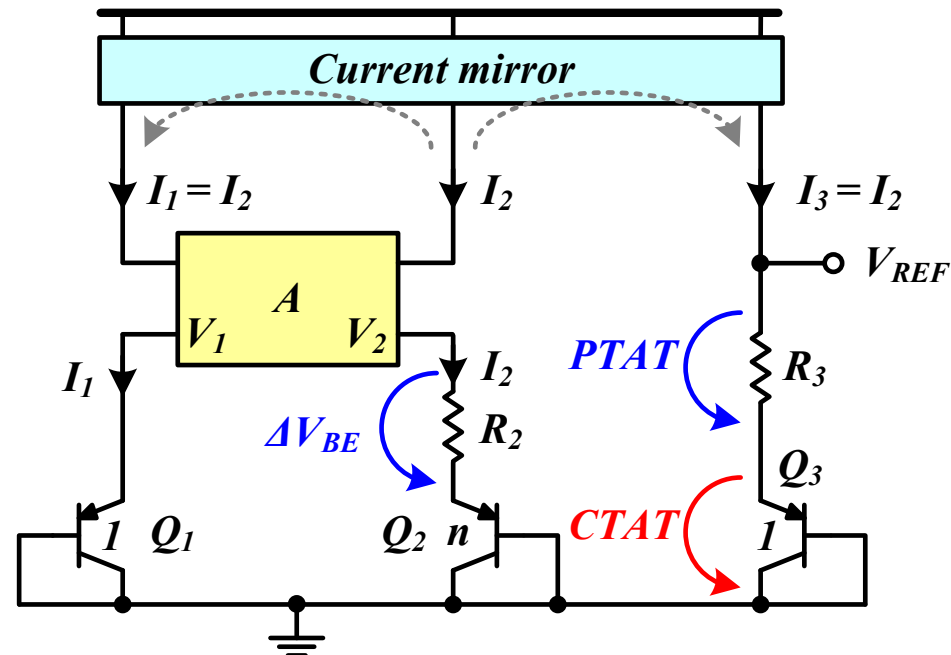
V_{BE} is CTAT

- It can be shown that $|V_{BE}| = V_T \ln \frac{I_C}{I_S} \approx V_{G0} - b_1 T \rightarrow$ CTAT
 - I_S is a strong function of temperature.
 - $|V_{BE}|$ is CTAT even if I_C itself is PTAT!
- All “simple” analytical models are inaccurate (and complicated).
 - Get b_1 from simulations: Usually $b_1 \approx 1.5 - 2 \text{ mV/K}$.



V_{REF} is ZTAT

- $V_{R_3} = I_3 R_3 = \Delta V_{BE} \cdot \frac{R_3}{R_2} = V_T \ln n \cdot \frac{R_3}{R_2} = a_1 T \rightarrow \text{PTAT}$
- $|V_{BE}| = V_T \ln \frac{I_C}{I_S} \approx V_{G0} - b_1 T \rightarrow \text{CTAT}$
- $V_{REF} = \Delta V_{BE} \cdot \frac{R_3}{R_2} + |V_{BE3}| = a_1 T + V_{G0} - b_1 T \approx V_{G0}$
- $V_{REF} = \text{PTAT} + \text{CTAT} = \text{ZTAT} \approx V_{G0}$



Why the Name “Bandgap”?

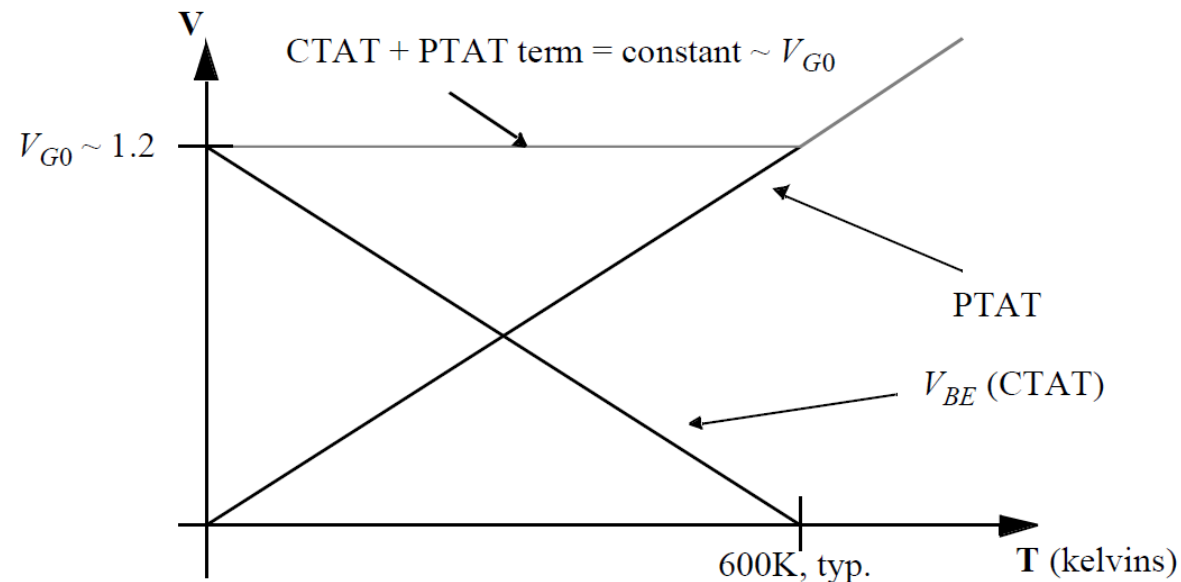
$$\square V_{R_3} = I_3 R_3 = \Delta V_{BE} \cdot \frac{R_3}{R_2} = V_T \ln n \cdot \frac{R_3}{R_2} = a_1 T \rightarrow \text{PTAT}$$

$$\square |V_{BE}| = V_T \ln \frac{I_C}{I_S} \approx V_{G0} - b_1 T \rightarrow \text{CTAT}$$

$$\square V_{REF} = \Delta V_{BE} \cdot \frac{R_3}{R_2} + |V_{BE3}| = a_1 T + V_{G0} - b_1 T \approx V_{G0}$$

$$\square V_{REF} = \text{PTAT} + \text{CTAT} = \text{ZTAT} \approx V_{G0}$$

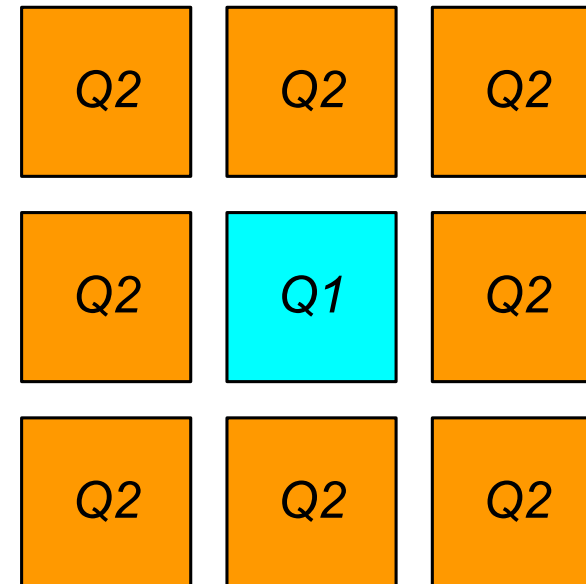
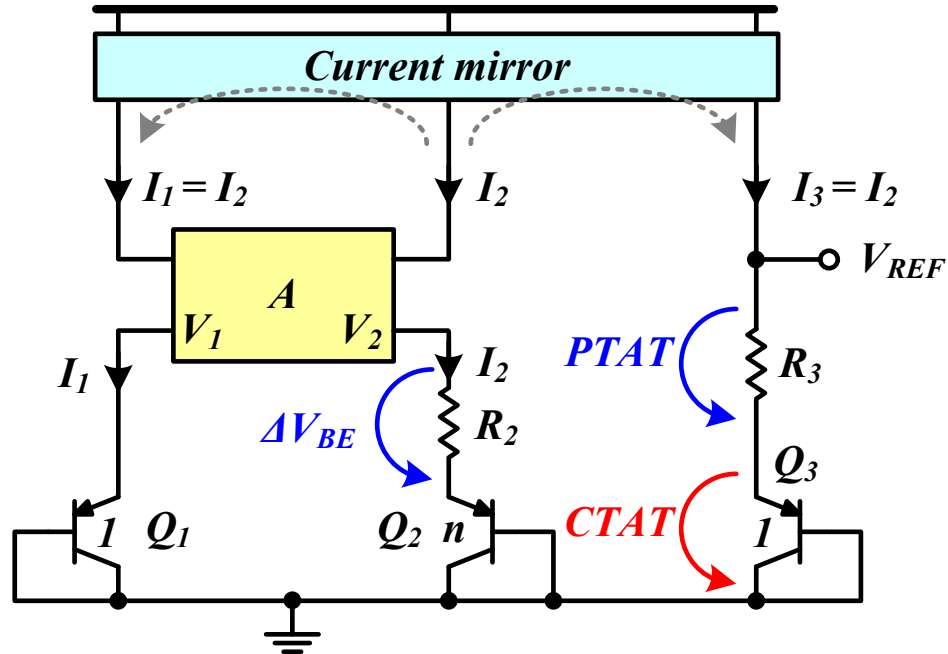
$V_{G0} = E_{g0}/q \approx 1.2 \text{ V}$ is the bandgap voltage (energy) of Si extrapolated at absolute zero Kelvin



BGR Design Example: n

□ Due to layout considerations, two values of n are usually used.

- $n = 8$
- $n = 24$



BGR Design Example: R_2

- Given current consumption select R_2 .

$$I_2 = \frac{\Delta V_{BE}}{R_2} = \frac{\frac{kT}{q} \ln n}{R_2}$$

- Ex: $I_{total} = 30\mu A$ and $n = 8$

$$I_2 = 10\mu A$$

$$R_2 \approx 5.4k\Omega$$

BGR Design Example: R_3

- ❑ Choose R_3 to achieve ZTAT V_{REF} (set $a_1 = b_1$).

$$V_{REF} = \Delta V_{BE} \cdot \frac{R_3}{R_2} + |V_{BE3}| = a_1 T + V_{G0} - b_1 T \approx V_{G0}$$

- ❑ $a_1 \approx \frac{k}{q} \ln n \cdot \frac{R_3}{R_2} \approx 0.086 \ln n \cdot \frac{R_3}{R_2} \text{ mV/K}$

- ❑ Get b_1 from simulations (all analytical models are inaccurate).

 - Usually $b_1 \approx 1.5 - 2 \text{ mV/K}$

- ❑ Ex: $n = 8$ and $b_1 = 1.8 \text{ mV/K}$

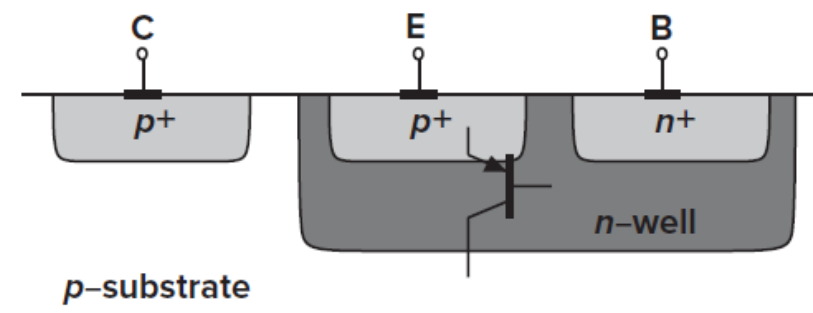
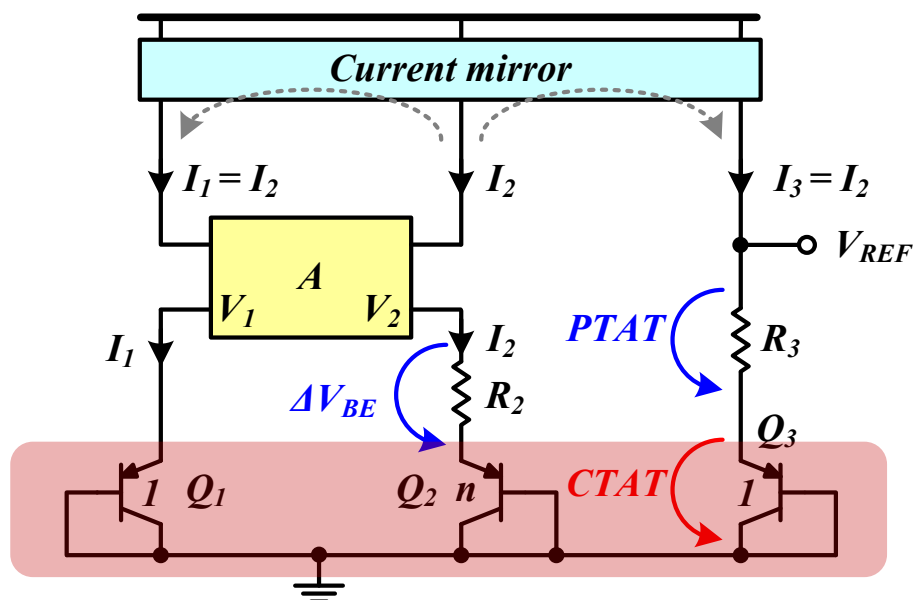
$$\frac{R_3}{R_2} \approx 10$$

$$R_3 \approx 54 \text{ k}\Omega$$

- ❑ Fine tune R_3 in SPICE.

Parasitic / Substrate / Native PNP

- ❑ The diode connected PNP can be implemented in any standard CMOS process.
 - But it usually has very low $\beta = I_C/I_B (< 10)$.
 - The n-well has relatively high parasitic ohmic resistance.
- ❑ Can be replaced by MOSFET in WI.
 - Necessary for low supply voltage operation.

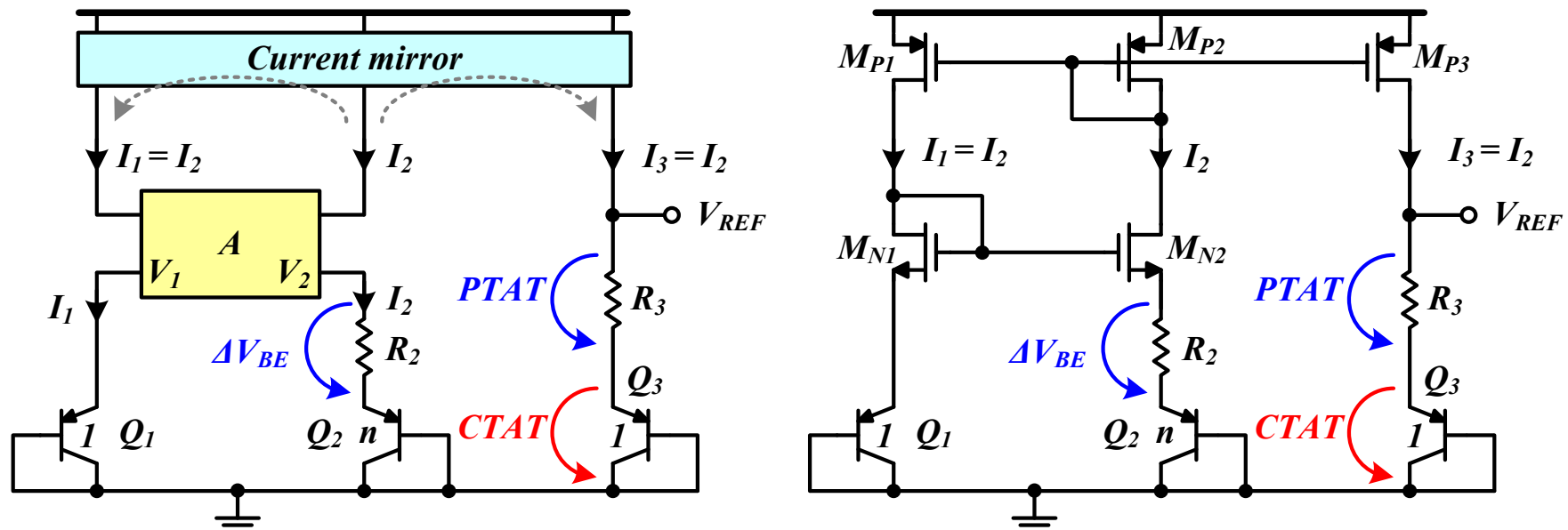


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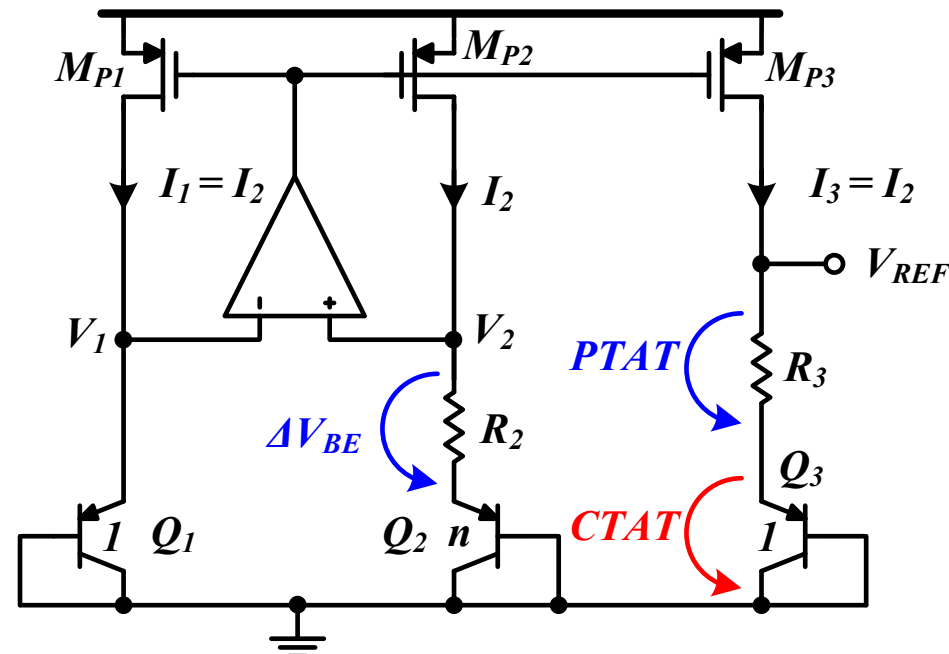
CMOS BGR Example (1)

- ❑ Current mirror: M_{P1-3}
- ❑ A-block: $M_{N1,2} \rightarrow$ Same current $\rightarrow$ Same V_{GS}
- ❑ Choice of L and W
 - Large L ($> 1\mu m$) is usually used: Reduce CLM and flicker noise.
 - For low supply voltage, bias the transistors in MI or WI
 - Given L , g_m/I_D , and $I_D \rightarrow$ get W from charts / look-up tables



CMOS BGR Example (2)

- ❑ The op-amp keeps V_1 and V_2 at the same voltage.
- ❑ The op-amp can be implemented as a simple 5T OTA.
 - Folded cascode may be used if wide input range is required.
- ❑ Bias the op-amp using a constant-gm circuit.
 - Or use the BGR itself to bias it (self-biased)!



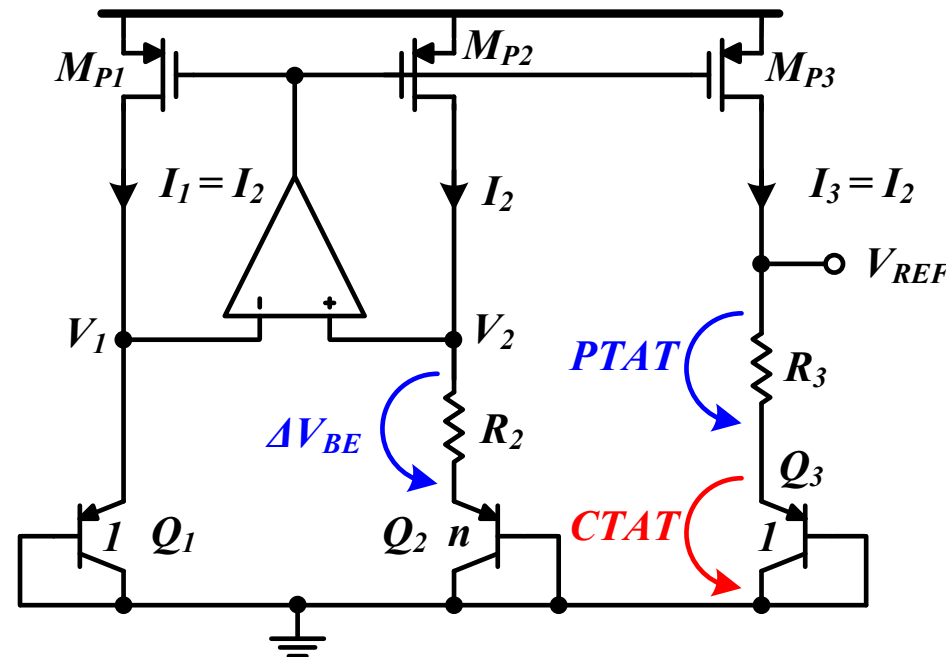
Positive or Negative FB?

- ❑ Cut the loop at OTA output: We must guarantee $\beta_N > \beta_P$.
- ❑ Note that $M_{P1,2}$ add 180° phase shift.

$$\beta_N \approx g_{mP2}(R_2 + 1/g_{mQ})$$

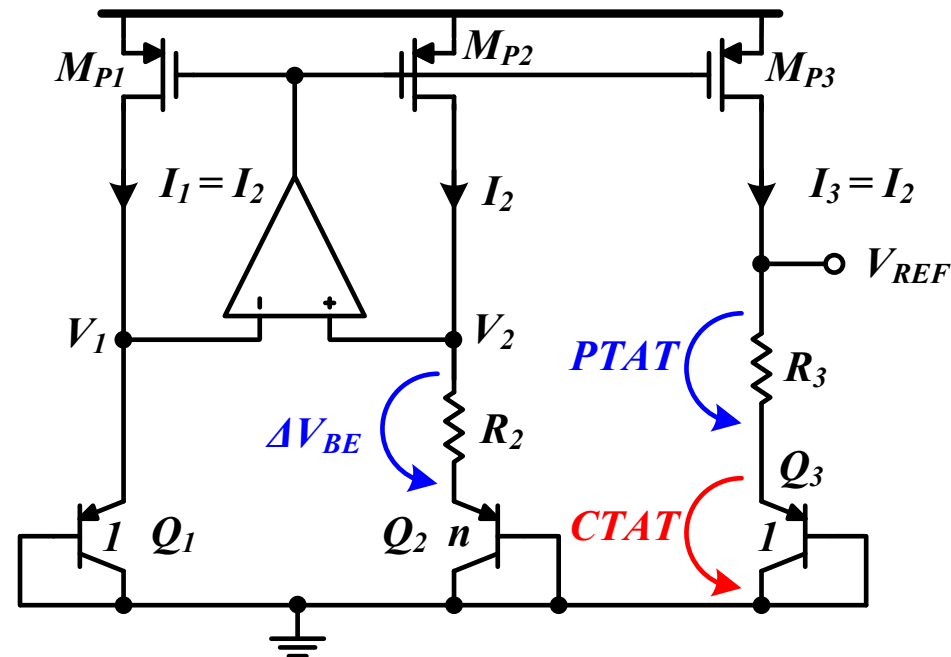
$$\beta_P \approx g_{mP1}(1/g_{mQ})$$

- ❑ Ex: To set $\beta_N > 2\beta_P \rightarrow R_2 > \frac{1}{g_m} = \frac{V_T}{I_C} \rightarrow V_{R_2} = V_T \ln n > V_T$



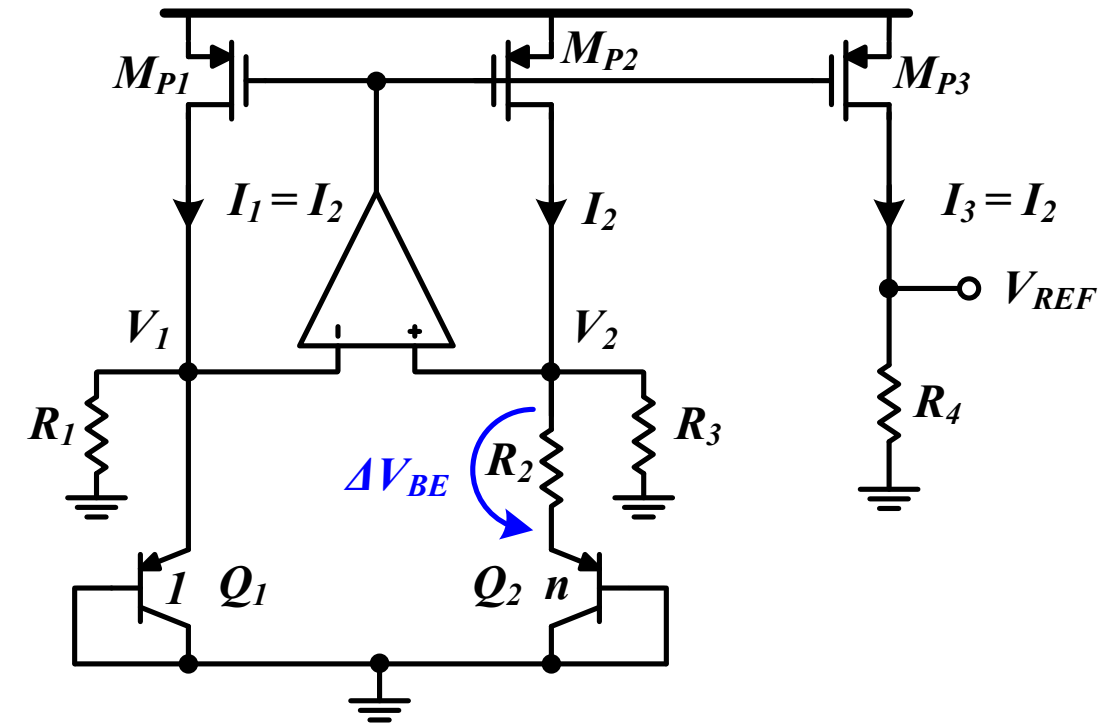
Low Voltage BGR: How?

- ❑ Adding PTAT and CTAT in voltage domain gives $V_{REF} \approx 1.2V$.
- ❑ For modern technologies, this value is higher than V_{DD} itself!
- ❑ The solution is to add PTAT and CTAT in the current domain.



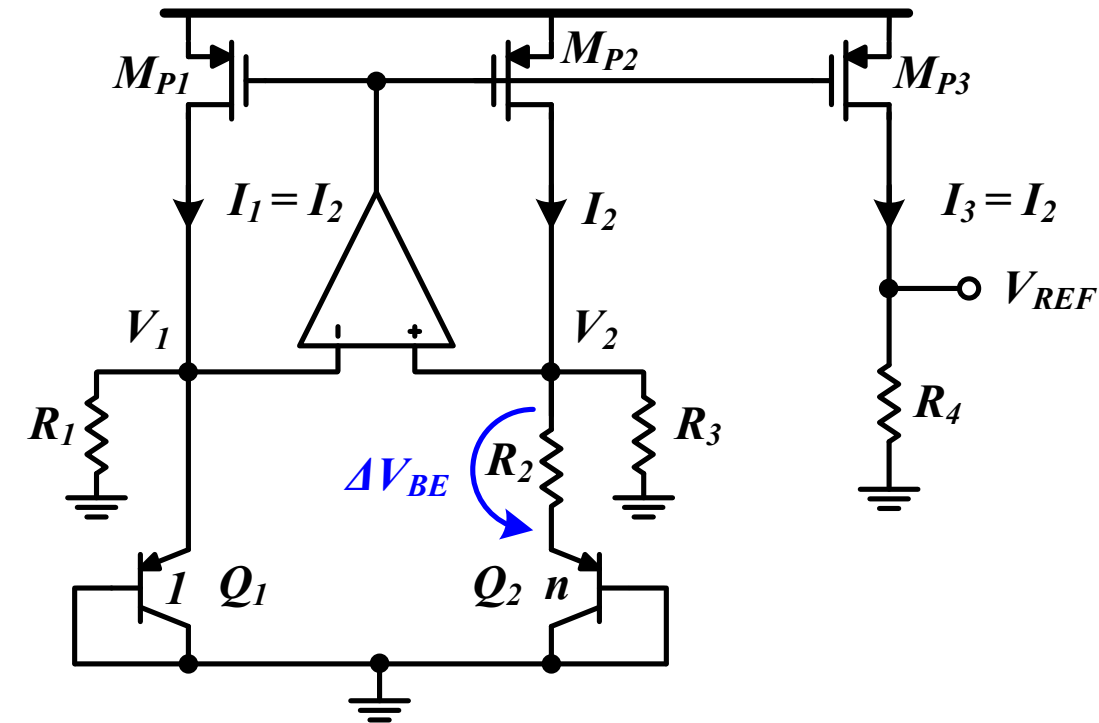
Low Voltage BGR

- ❑ Q2 carries PTAT current
- ❑ R_3 converts CTAT voltage (V_{BE1}) to CTAT current.
- ❑ $R_1 = R_3$ maintains $I_{E1} = I_{E2}$
- ❑ I_2 is PTAT + CTAT = ZTAT
- ❑ $V_{REF} = R_4 \left(\frac{\Delta V_{BE}}{R_2} + \frac{V_{BE1}}{R_3} \right) < 1.2V$



Low Voltage BGR: Design Procedure

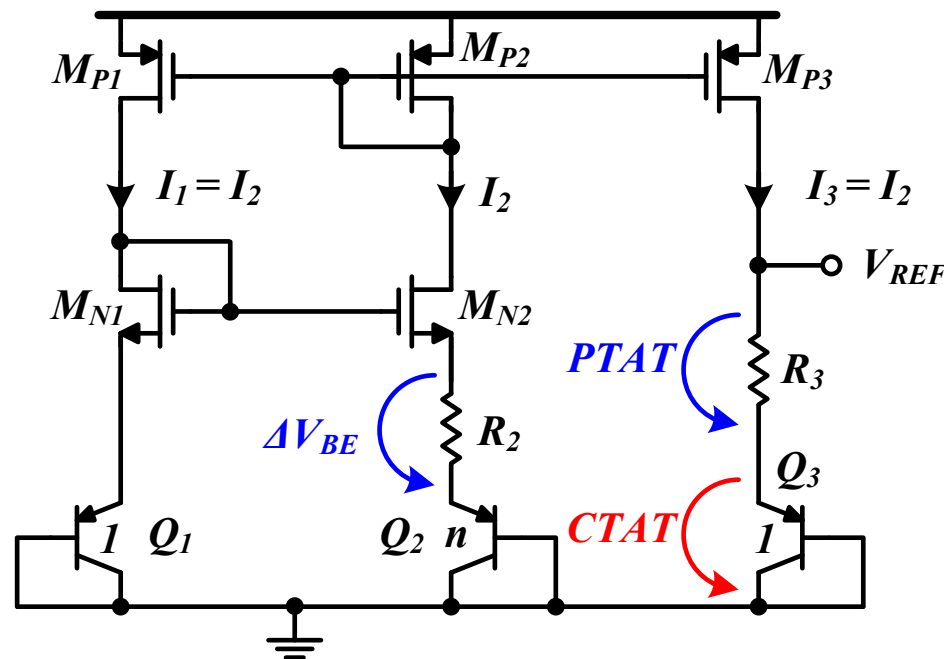
- $V_{REF} = R_4 \left(\frac{\Delta V_{BE}}{R_2} + \frac{V_{BE1}}{R_3} \right) < 1.2V$
- $V_{REF} = R_4 \left(\frac{a_1 T}{R_2} + \frac{V_{G0} - b_1 T}{R_3} \right) \approx V_{G0} \cdot \frac{R_4}{R_3}$
- $\frac{\partial V_{REF}}{\partial T} = 0 \Rightarrow \frac{a_1}{R_2} - \frac{b_1}{R_3} = 0$
 - $a_1 \approx \frac{k}{q} \ln n$
 - $b_1 \approx 1.5 - 2 \text{ mV/K}$
- Design procedure:



- Assume $I_{R_2} \rightarrow$ Given ΔV_{BE} find $R_2 \rightarrow$ Calc V_2 and $R_3 \rightarrow I_2 \rightarrow$ Given V_{REF} find R_4

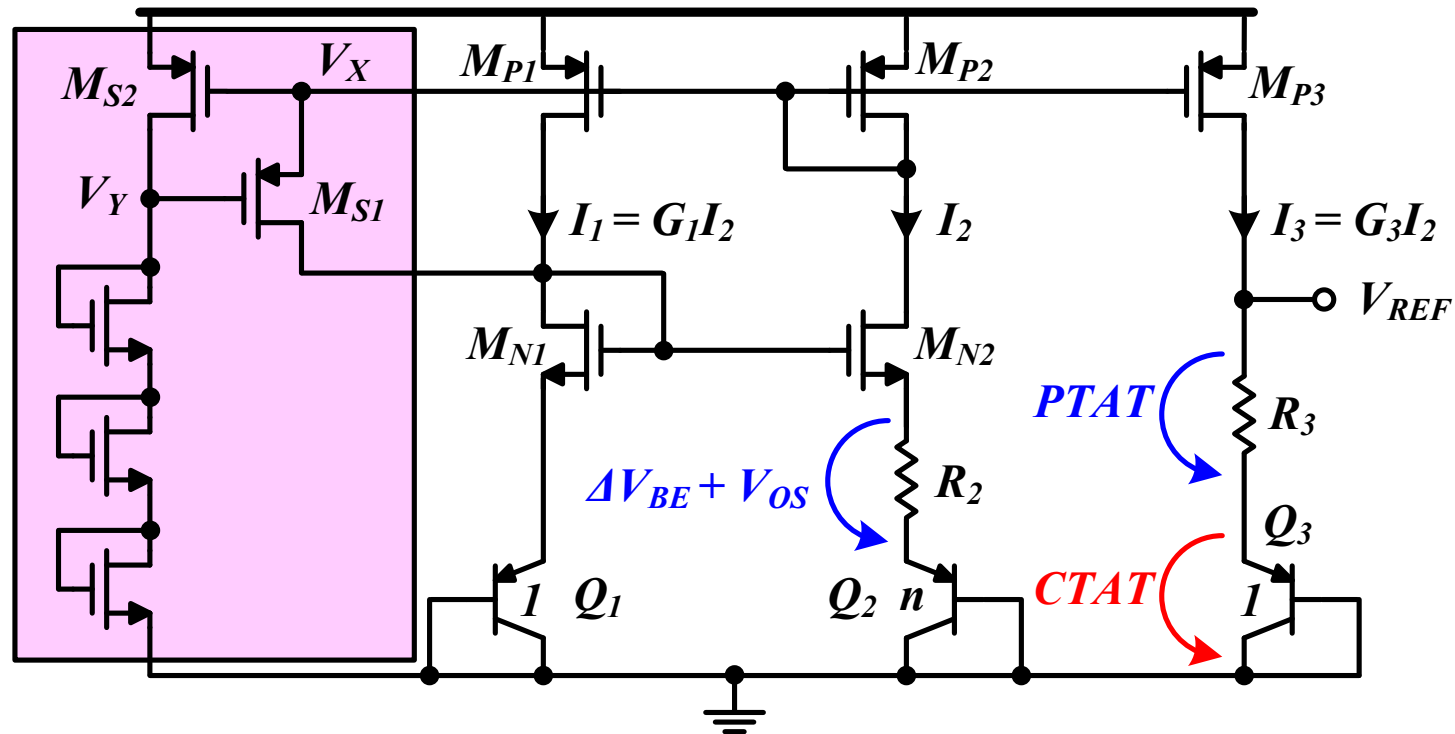
The Start-up Problem

- ❑ All currents = 0 is another valid solution for the circuit!
- ❑ Start-up circuit must drive the circuit out of the zero bias point.
 - Then it should automatically turn-off or consume little current
- ❑ Start-up verified by ramping-up VDD from zero in DC sweep and transient simulations.



Startup Circuit Example

- Startup problem means $V_X = V_{DD}$ and $V_Y = 0$.
 - M_{S1} will turn on charging the gates of $M_{N1,2}$.
 - $Q_{1,2}$ will turn on and the BGR starts.
 - V_Y will increase turning off M_{S1} and driving M_{S2} in linear region.



BGR Curvature

- PTAT and CTAT are not perfectly linear.
 - They both have convex upward curvature → Curvatures add!
- If expressed mathematically:

$$V_{PTAT} \approx a_1 T - a_2 T^2$$

$$V_{CTAT} \approx V_{G0} - b_1 T - b_2 T^2$$

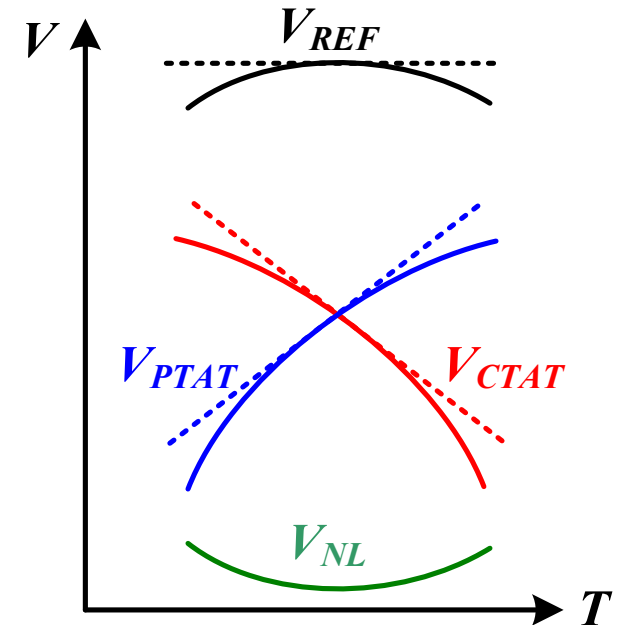
$$V_{REF} \approx V_{G0} + (a_1 - b_1)T - (a_2 + b_2)T^2$$

- We can set $a_1 = b_1$
- But a_2 and b_2 add (convex upward)

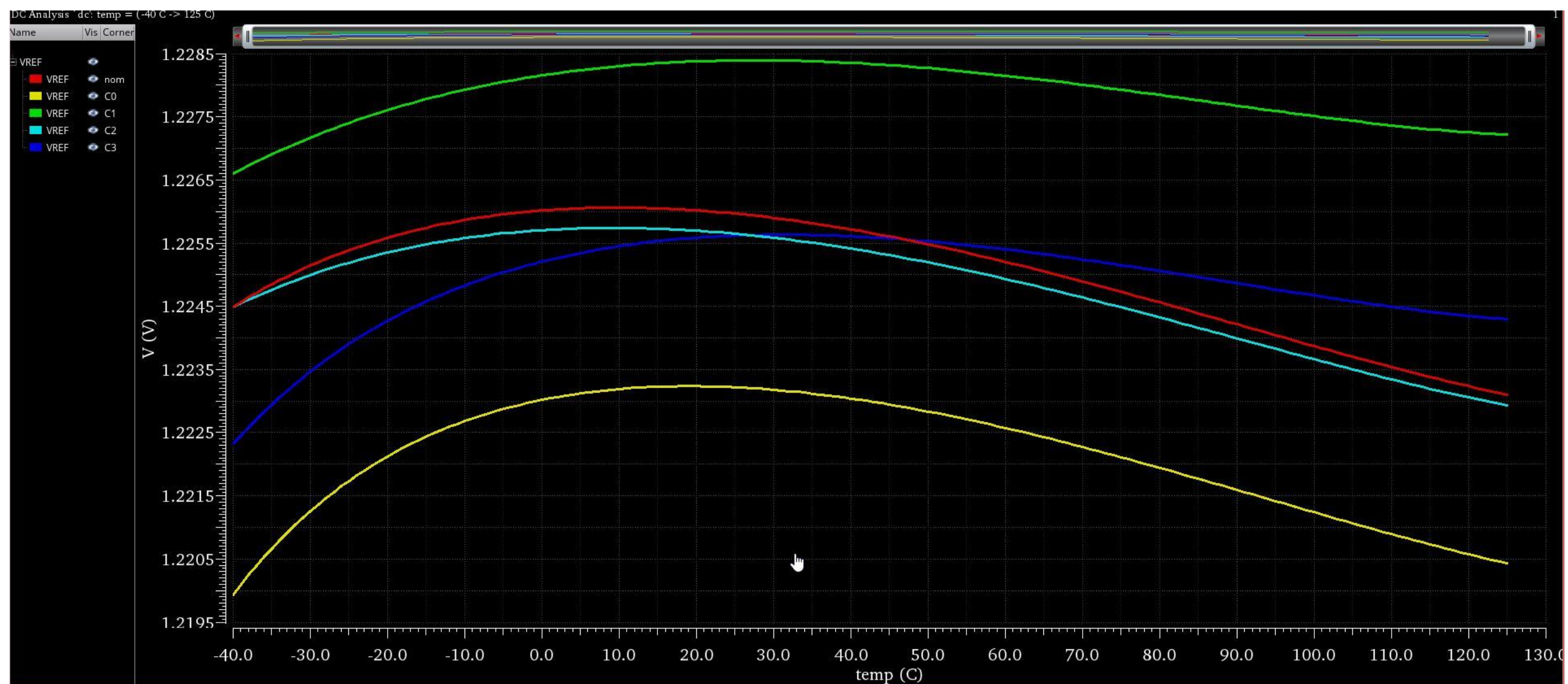
$$V_{REF} \approx V_{G0} - (a_2 + b_2)T^2$$

- Curvature can be corrected by adding a convex downward signal (e.g., $PTAT^2$).

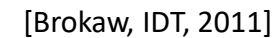
$$V_{NL} \approx c_2 T^2 \rightarrow c_2 = a_2 + b_2$$



Example of BGR Simulation Results



23: Reference Circuits

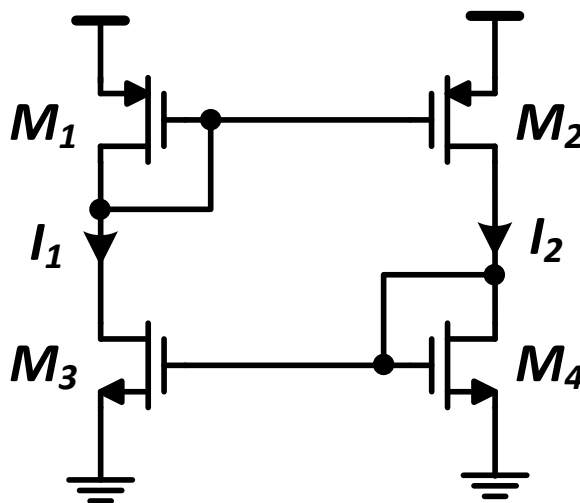


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Supply Independent Reference Current

- ❑ To avoid V_{DD} dependence, the circuit must bias itself!
- ❑ The circuit is governed by only one equation: $I_1 = I_2$
- ❑ The currents are supply-independent (if CLM is neglected)
 - But they are undefined!
 - To uniquely define the currents, we must add another constraint to the circuit



Self-Biased Circuit

- Assume sq. law is valid:

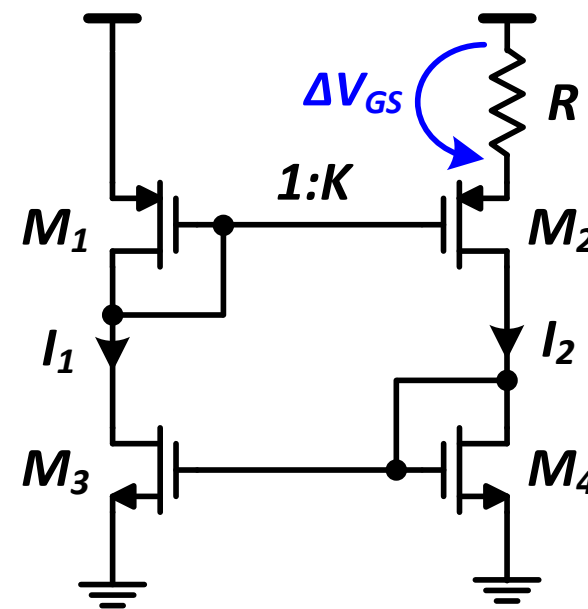
$$I_D = \frac{\beta}{2} (V_{GS} - V_{TH})^2$$

$$\frac{\beta_2}{\beta_1} = \frac{W_2}{W_1} = K$$

$$\Delta V_{GS} = \sqrt{\frac{2I_{1,2}}{\beta_1}} - \sqrt{\frac{2I_{1,2}}{K\beta_1}} = I_{1,2}R$$

- Solve quadratic equation for $\sqrt{I_{1,2}}$

- 1st soln: $\sqrt{I_{1,2}} = 0$
 - Startup circuit is required!
- 2nd soln: $\sqrt{I_{1,2}} = \frac{1}{R} \sqrt{\frac{2}{\beta_1}} \left(1 - \frac{1}{\sqrt{K}}\right)$
 - Is this useful?



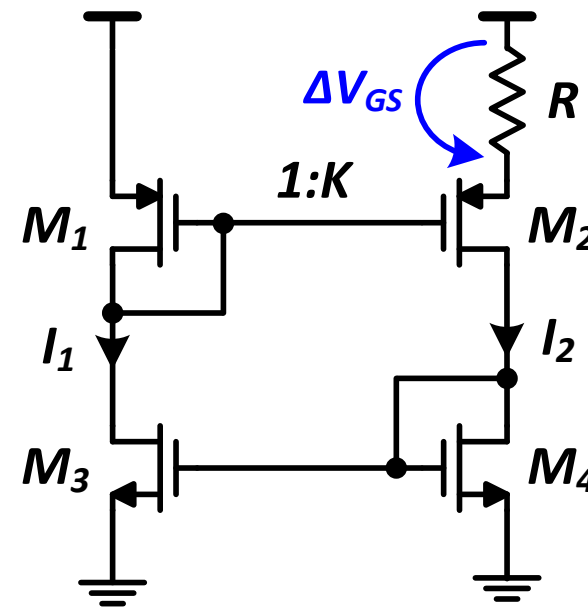
Self-Biased Circuit

$$I_{1,2} = \frac{1}{R^2} \cdot \frac{2}{\beta_1} \left(1 - \frac{1}{\sqrt{K}} \right)^2$$

- $I_{1,2}$ independent of supply, but depends on process and temp

$$g_{m1} = \sqrt{\beta_1 \cdot 2I_1} = \frac{2}{R} \left(1 - \frac{1}{\sqrt{K}} \right)$$

- Independent of process, supply, and temperature
 - Assuming ideal R



Constant- g_m Circuit

$$g_{m1} = \sqrt{\beta_1 \cdot 2I_1} = \frac{2}{R} \left(1 - \frac{1}{\sqrt{K}} \right)$$

□ Independent of process, supply, and temperature

- Assuming ideal R

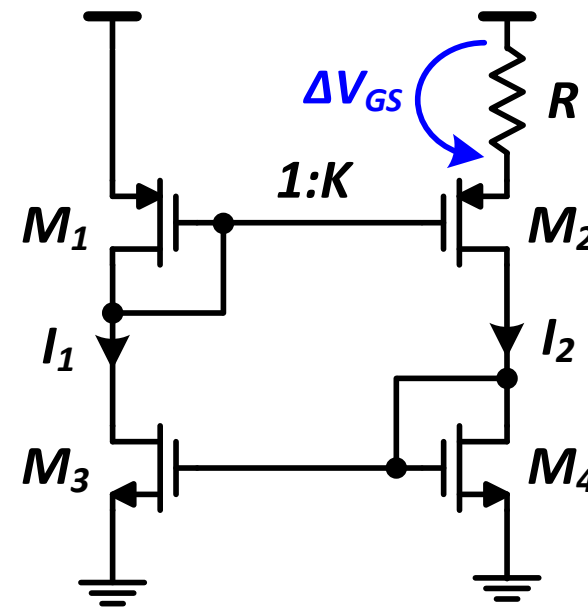
□ g_m is what actually matters!

- Determines gain, noise, and speed.
- We want constant- g_m !

□ Ex: $K = 4 \rightarrow g_{m1} = \frac{1}{R}$

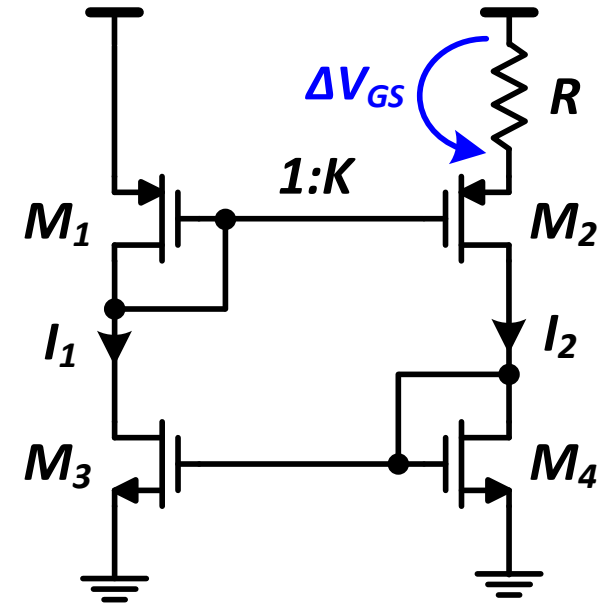
- Let $|A_v| = g_{m1}R_D = \frac{R_D}{R} = \text{constant!}$

□ It can be shown that g_m is still constant even if biased at MI or WI!



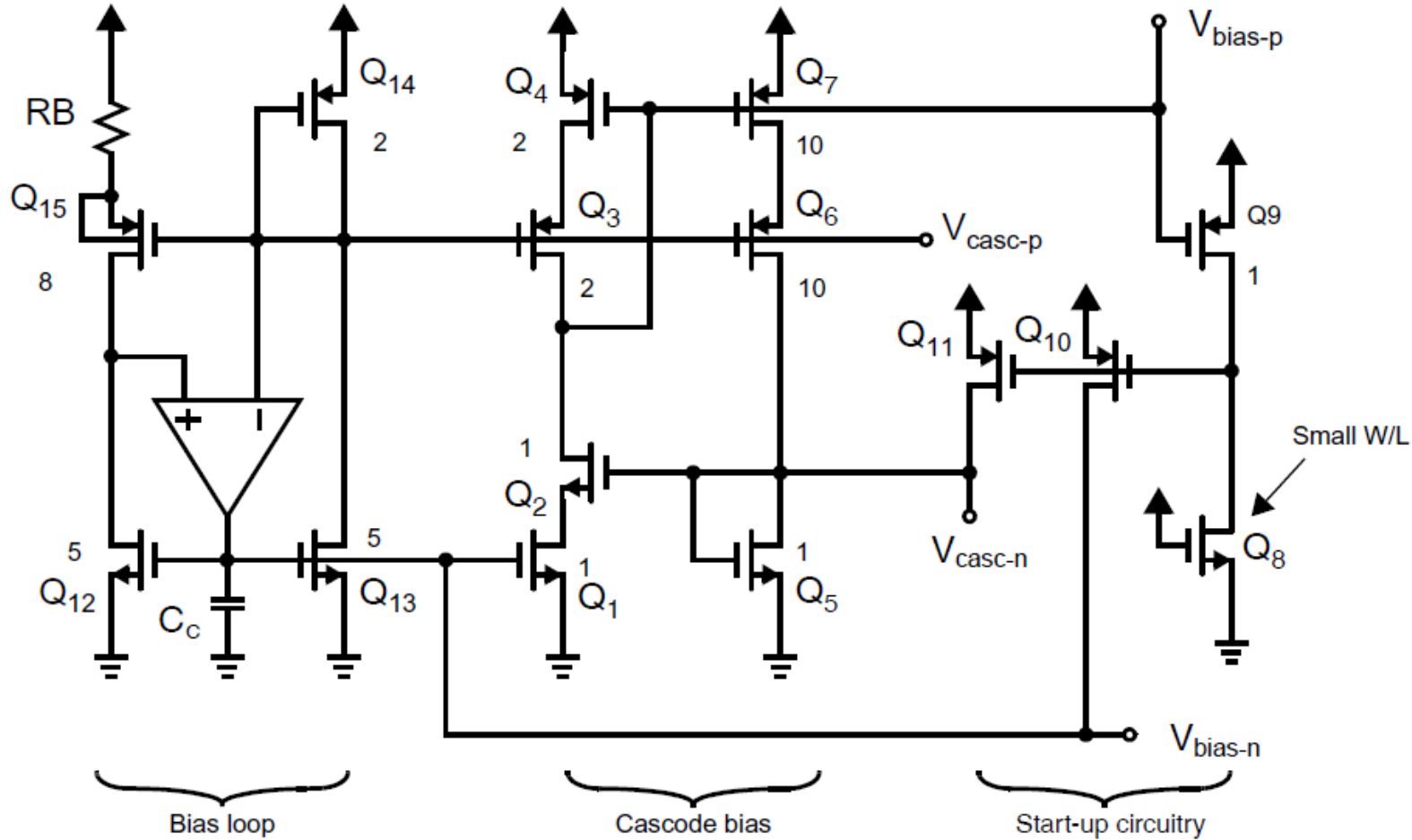
Quiz

- Note that the FB loop formed by M1-M4 is a +ve FB loop.
 - Does it oscillate?
 - Find the loop gain and find the stability condition ($LG < 1$)
 - Is the stability affected if R is off-chip?



OTA Bias Circuit Example

- $Q_{12,13} \rightarrow V_{bias-n} \rightarrow Q_1 \rightarrow$
 $Q_4 \rightarrow V_{bias-p}$
- $Q_4 \rightarrow Q_7 \rightarrow Q_5 \rightarrow V_{casc-n}$
 $(I_{Q5} = 5 \times I_{Q1})$
- $Q_{14} \rightarrow V_{casc-p} (I_{Q14} = 5 \times$
 $I_{Q4})$
- Startup: Q_8 (large res) $\rightarrow$
 $Q_{10,11}$ (ON) $\rightarrow Q_{5,13} \rightarrow Q_9 \rightarrow$
 $Q_{10,11}$ (OFF)



MOSFET Temperature Effects

- V_{TH} of long channel MOS:

$$\partial V_{THN}/\partial T \approx -1 \text{ mV/K and } \partial V_{THP}/\partial T \approx -1.4 \text{ mV/K}$$

- V_{TH} of short channel MOS:

$$\partial V_{THN}/\partial T \approx -0.6 \text{ mV/K and } \partial V_{THP}/\partial T \approx -0.6 \text{ mV/K}$$

- Mobility: $\mu \approx \mu(T_o) \left(\frac{T_o}{T}\right)^{1.5}$

- V_{TH} dominate at low V_{GS} (WI): $I_D \uparrow$ with temperature

- μ dominate at high V_{GS} (SI): $I_D \downarrow$ with temperature

Thank you!

References

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